

BizRadar v4

Universal Algorithm Specification

Batch 10 — Classifier Expansion to 100% Coverage

WHAT THIS BATCH DELIVERS

Takes BizRadar's classifier from partial coverage to 100% of the 1,464 business types in the Master Reference. Every NAICS-2 sector is classified by evidence strength (STRONG / MODERATE / THIN / GAP), every Part-2 modern format (food trucks, ghost kitchens, lash bars, escape rooms, etc.) is mapped to its NAICS parent and scoped IN or OUT of BizRadar's discovery surface. The Layer 0 input classifier is upgraded from 6 modes to 10, with explicit disambiguation rules for the most common ambiguous queries ('bakery', 'studio', 'spa', 'salon', 'shop', 'agency'). Ships a named out-of-scope filter so the classifier refuses NAICS-21/22/55/92 inputs cleanly rather than guessing.

WHAT THIS BATCH IS NOT

(1) An implementation of a new classifier — Batch 10 is spec-only. The Layer 0 changes ship as code in Batch 12. (2) A completion of every profile template — the spec defines which templates need writing, not the templates themselves (those land in Batch 11). (3) A research expansion — Batch 9 already ground-truthed the customer-facing claims; this batch is purely structural.

Batch	10 of 12
Total inventory	1,464 business types (1,012 NAICS-6 + 452 Part-2 modern)
NAICS-2 sectors covered	20 of 20 (full structural coverage)
Coverage classes	STRONG (NAICS-72 fully; 713940 gym fully); MODERATE (~10 sectors); THIN (~5 sectors); GAP (~2 sectors out of scope)
Layer 0 modes (current → spec)	6 → 10
Disambiguation rules	6 worked examples; full set in registry
Out-of-scope filter	4 NAICS-2 sectors + 6 NAICS-6 patterns named
Dependencies	Batch 8 (TAM/survival), Batch 9 (verified studies)
Pages	~80 (target hit)

Executive Summary

Through Batch 9 BizRadar's spec layer was strong on the *narrative* dimension — every customer-facing claim now ties back to a verified study — but the *classifier* dimension was still uneven. Some NAICS-2 sectors had complete profile templates and grounded recommendations (NAICS-72: Accommodation & Food Services); others had nothing concrete (NAICS-23: Construction, despite being one of the largest sectors by establishment count). Batch 10 closes that unevenness.

Coverage at a glance

Coverage class	NAICS-2 sectors	NAICS-6 codes	% of 1,012
STRONG (direct sector evidence + Batch 9 studies)	3	97	9.6%
MODERATE (cross-industry evidence; Batch 9 cross-sector studies)	10	404	39.9%
THIN (B2B-leaning; little BizRadar fit)	6	482	47.6%
GAP (out of scope)	1	29	2.9%

What changed since Batch 9

Batch 9 was content-dense (20 study chapters, 60-page deliverable). Batch 10 is structurally dense — it covers more business types but the per-type detail is lighter. The two batches together establish BizRadar's spec footprint: Batch 9 is the *what* (what the evidence says); Batch 10 is the *where* (which businesses the evidence applies to).

Headline structural decisions

1. 100% NAICS-2 coverage. All 20 sectors are in the spec — even those scoped out (NAICS-21, 22, 55, 92). Out-of-scope sectors are EXPLICITLY out of scope, not silently absent. Classifier must respond with a clear out-of-scope flag.
2. Coverage classified by evidence strength, not by NAICS-2 size. NAICS-31-33 (Manufacturing) has 346 codes — the largest sector — but is THIN coverage because most manufacturing is B2B and outside BizRadar's local-discovery surface. NAICS-72 has 15 codes — the smallest sector by NAICS-6 count among in-scope sectors — but is STRONG coverage because it's where BizRadar's verified evidence lands.
3. Modern formats (Part 2's 452 entries) are first-class. The Layer 0 classifier is upgraded to recognize them by name (food truck, ghost kitchen, boba shop, lash bar, etc.) and route them to their closest NAICS parent.
4. Disambiguation is explicit. The six common ambiguous queries (bakery, studio, spa, salon, shop, agency) have named tiebreak rules. The classifier no longer guesses — it either applies a deterministic rule or asks the user.
5. Chains are flagged. The Luca-Harvard finding from Batch 9 — that Yelp star ratings do not move revenue at chain restaurants — has implications across BizRadar's whole review-management narrative. Top-100 chain detection is spec'd as part of Layer 0; chain templates differ from independent templates.

Methodology — Coverage Audit

The coverage audit takes the Master Reference's 1,464 entries and grades each NAICS-2 sector against three evidence layers from earlier batches:

Three evidence layers

Layer	Source	What it provides	Earlier batch
L1	Census County Business Patterns 2023	Establishment count per NAICS-6 (TAM proxy).	Batch 8
L2	BLS Business Employment Dynamics (BED) 2013-cohort survival	5-year survival rate by NAICS-2 (failure-rate input for risk callouts).	Batch 8
L3	Verified studies (Cornell, HBS, HFA, McKinsey, Womply, etc.)	Customer-facing claim grounding (specific magnitudes for review/retention/CX).	Batch 9

Coverage class definitions

Each NAICS-2 sector is classified into one of four coverage classes based on which evidence layers apply:

Class	L1	L2	L3	BizRadar treatment
STRONG	✓	✓	Direct sector studies	Full reports; sector-specific magnitudes.
MODERATE	✓	✓	Cross-industry studies only	Reports with cross-sector magnitudes; flag as such.
THIN	✓	✓	Cross-industry studies only AND BizRadar's discovery surface is weak (B2B, regulated, manufacturing)	Limited reports; recommend the user verify BizRadar fit.
GAP	—	—	Out of scope by design	Reject input; out-of-scope flag.

ON THE MODERATE/STRONG DISTINCTION

The boundary between MODERATE and STRONG is whether at least one Batch-9 verified study is sector-specific (not just cross-industry). NAICS-72 has S001-S007 (7 hotel/restaurant studies). NAICS-44/45 has S014 (Womply Retail). NAICS-71 has S008/S009 for the gym sub-sector (713940) but cross-industry-only for the rest of arts/recreation. Sectors with just S010-S020 (cross-industry only) are MODERATE, not STRONG.

Why this taxonomy

- **Predictable customer expectations.** A customer running BizRadar on a hotel gets sector-specific RevPAR-elasticity estimates from Cornell. A customer running BizRadar on a plumber gets cross-industry retention/review numbers with explicit 'this is cross-sector — sector-specific work is a research gap' framing. Both deserve a confident answer; both deserve clear scoping.
- **Clear research roadmap.** The MODERATE-to-STRONG promotion is the unit of work for Batch 11 sector polish. Construction (NAICS-23), Health Care (NAICS-62), and Personal Services (NAICS-81) are the highest-priority MODERATE sectors that would benefit from sector-specific studies.
- **Explicit out-of-scope.** Manufacturing (NAICS 31-33), Mining (NAICS-21), and Public Administration (NAICS-92) are OUT — silently. The classifier must surface this explicitly so users don't get a low-quality report for a sector BizRadar isn't designed to serve.

Coverage Taxonomy — All 20 NAICS-2 Sectors

100% of the 1,464 Master Reference entries are accounted for. Every sector is named; nothing is silently excluded.

The grid below is the master coverage table — referenced throughout the rest of this batch and pulled into the renderer as the source of truth for which profile templates exist, which need writing in Batch 11, and which the classifier should refuse to handle entirely.

NAICS	Sector	# codes	Class	Priority
11	Agriculture, Forestry, Fishing & Hunting	64	MODERATE	Low
21	Mining, Quarrying, Oil & Gas Extraction	21	THIN	Defer
22	Utilities	14	THIN	Defer
23	Construction	31	MODERATE	HIGH
31-33	Manufacturing	346	THIN	MEDIUM
42	Wholesale Trade	69	THIN	Defer
44-45	Retail Trade	57	STRONG	HIGH
48-49	Transportation & Warehousing	57	MODERATE	MEDIUM
51	Information	29	THIN	Defer
52	Finance & Insurance	35	MODERATE	MEDIUM
53	Real Estate, Rental & Leasing	24	MODERATE	MEDIUM
54	Professional, Scientific & Technical Services	49	MODERATE	HIGH
55	Management of Companies and Enterprises	3	THIN	Defer permanently
56	Administrative, Support & Waste Management	44	MODERATE	HIGH
61	Educational Services	17	MODERATE	MEDIUM
62	Health Care & Social Assistance	39	MODERATE	HIGH
71	Arts, Entertainment & Recreation	25	STRONG	HIGH for non-g
72	Accommodation & Food Services	15	STRONG	DONE
81	Other Services (except Public Admin)	44	MODERATE	HIGH
92	Public Administration	29	GAP	Defer permanently

READING THIS GRID

Sectors marked HIGH priority are the ones Batch 11 will work on directly: Retail (44-45), Construction (23), Professional Services (54), Other Services (81), Administrative Services (56), Health Care (62), Arts/Recreation (71). These are the sectors with the most BizRadar customer demand AND the most room for sector-specific verified-studies expansion. Sectors marked Defer/THIN/GAP are explicitly out — the classifier should not produce a customer report for them.

Sector-by-Sector Deep Dives

Each NAICS-2 sector gets its own chapter. Same structure: evidence-layer status (L1 TAM, L2 survival, L3 verified studies), coverage notes, BizRadar priority. The 20 chapters below cover every dollar of NAICS-classified U.S. economic activity — including the sectors BizRadar explicitly does NOT serve.

HOW TO USE THESE CHAPTERS

Use them as the source of truth when adding new profile templates in Batch 11 or when answering 'why isn't BizRadar covering my industry?' questions. Each chapter ends with a named priority — HIGH (write templates now), MEDIUM (write in a future batch), Defer (out of scope).

NAICS 11 — Agriculture, Forestry, Fishing & Hunting

Coverage class: MODERATE | 64 NAICS-6 codes | Priority: Low

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 51% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only (S010, S017, S020)

Coverage notes

BizRadar's customer base in agriculture is thin in v3. Direct-to-consumer models (CSAs, farmstands, U-pick) are the most actionable subset. Most NAICS-6 codes here are primary-production and not BizRadar's natural fit.

BIZRADAR PRIORITY

Low — primary-production codes outside BizRadar's discovery surface (Google Places, Yelp).

NAICS 21 — Mining, Quarrying, Oil & Gas Extraction

Coverage class: THIN | 21 NAICS-6 codes | Priority: Defer

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 56% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only (S017)

Coverage notes

Out of BizRadar's natural scope. Mining and extraction are heavy-capex sectors with sales pipelines that don't show up in local-search reviews. Coverage is structural (the codes exist) but BizRadar is unlikely to ever serve a customer in NAICS-21.

BIZRADAR PRIORITY

Defer — flag in classifier as 'out of scope' if user submits a 21-* business.

NAICS 22 — Utilities

Coverage class: THIN | 14 NAICS-6 codes | Priority: Defer

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 65% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

Regulated utilities are out of scope. The only BizRadar-relevant subset is small water/sewage haulers (NAICS 221310 partial) — too thin to specialize.

BIZRADAR PRIORITY

Defer — out-of-scope flag in classifier.

NAICS 23 — Construction

Coverage class: MODERATE | 31 NAICS-6 codes | Priority: HIGH

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — ~750K establishments
L2 — BLS BED 2013-cohort survival	BLS BED — 36% 5-yr survival (one of the lowest)
L3 — Batch 9 verified studies	Cross-industry only (S017, S020) + indirect S006

Coverage notes

High-volume, low-margin sector with strong reviews presence on Google Maps for residential trades (plumbing, HVAC, electrical, painting). The 36% survival rate is one of the harshest in the BLS data — a critical risk-flag for BizRadar reports. Direct sector-specific peer-reviewed work on construction review-economics is a research gap.

BIZRADAR PRIORITY

HIGH — Construction is a top-3 BizRadar opportunity by establishment count. Profile templates needed for (at minimum) residential general contractors, plumbing, HVAC, electrical, roofing, landscaping.

NAICS 31-33 — Manufacturing

Coverage class: THIN | 346 NAICS-6 codes | Priority: MEDIUM

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 49% 5-yr survival (mfg average)
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

346 codes — by FAR the largest sector by NAICS-6 count. But ~95% are B2B and don't have a local-discovery footprint. The BizRadar-relevant subset is craft food & beverage manufacturing (311811 retail bakeries, 312120 breweries, 312130 wineries, 312140 distilleries) and small-batch consumer goods (e.g. 31523 cut & sew apparel).

BIZRADAR PRIORITY

MEDIUM — restrict coverage to <30 consumer-facing NAICS-6 codes within the sector. The other ~316 codes are out-of-scope by design.

NAICS 42 — Wholesale Trade

Coverage class: THIN | 69 NAICS-6 codes | Priority: Defer

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 53% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

B2B wholesale is out of BizRadar's local-discovery scope. The exception is wholesale florists (424930), wholesale auto-parts (423120), and a few other edge-cases that do have foot-traffic. <5 of 69 codes.

BIZRADAR PRIORITY

Defer — out-of-scope flag for 42-* unless user explicitly opts into B2B coverage in a future batch.

NAICS 44-45 — Retail Trade

Coverage class: STRONG | 57 NAICS-6 codes | Priority: HIGH

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — ~1M establishments
L2 — BLS BED 2013-cohort survival	BLS BED — 47% 5-yr survival
L3 — Batch 9 verified studies	S014 (Womply Retail), S010-S012 (cross-industry review economics), S015 (BrightLocal), S017 (Reichheld retention)

Coverage notes

Direct sector evidence for review-volume thresholds (S014: 43-review threshold, +26%; 200+ reviews, +46%). Retail is a STRONG-coverage sector. All 57 NAICS-6 codes warrant a profile template. Subsector specialization is needed for grocery (445*), apparel (458*), electronics (449210), and sporting goods (459110) given their distinct customer-acquisition models.

BIZRADAR PRIORITY

HIGH — top-3 BizRadar sector. Already partially covered through Batches 5-7; expand to all 57 codes.

NAICS 48-49 — Transportation & Warehousing

Coverage class: MODERATE | 57 NAICS-6 codes | Priority: MEDIUM

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 47% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

BizRadar-relevant subset: courier/local delivery (492210), taxi/limo (485310, 485320), moving services (484210), self-storage (493190 — actually under 53). These DO have reviews. ~10 of 57 codes are in scope; the rest (rail, air freight, pipeline) are not.

BIZRADAR PRIORITY

MEDIUM — restrict to local-courier and passenger-ground-transport subset.

NAICS 51 — Information

Coverage class: THIN | 29 NAICS-6 codes | Priority: Defer

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 49% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

Mostly software publishers, telecom carriers, broadcast — not BizRadar's natural fit. The exception: small publishers, podcasters, and content-creator businesses (519290) which have brand-building review surfaces (though not Google-Places-style).

BIZRADAR PRIORITY

Defer — out-of-scope flag for most of the sector.

NAICS 52 — Finance & Insurance

Coverage class: MODERATE | 35 NAICS-6 codes | Priority: MEDIUM

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 51% 5-yr survival
L3 — Batch 9 verified studies	S017 retention; partial fit for SMB advisory (referenced from RIA Schwab data — see Batch 9 follow-ups for verification).

Coverage notes

Sub-sector in scope: independent insurance agents (524210), tax-prep services (541213 — actually classified under 54), small RIAs (523930). 97% client-retention figure for RIAs from Batch 9 follow-ups (not in S001-S020 main set). Banks and credit unions out of scope.

BIZRADAR PRIORITY

MEDIUM — restrict to insurance agents, RIA, tax-prep.

NAICS 53 — Real Estate, Rental & Leasing

Coverage class: MODERATE | 24 NAICS-6 codes | Priority: MEDIUM

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 50% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only (S015 reviews, S020 trust)

Coverage notes

Real estate brokerages (531210), property managers (531311), self-storage (531130) all have rich review footprints. Equipment rental (532*) less so. Roughly 12 of 24 codes are in scope.

BIZRADAR PRIORITY

MEDIUM — covered partially in Batches 5-7. Need RE-broker template + self-storage template.

NAICS 54 — Professional, Scientific & Technical Services

Coverage class: MODERATE | 49 NAICS-6 codes | Priority: HIGH

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — ~1.4M establishments
L2 — BLS BED 2013-cohort survival	BLS BED — 51% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only — no sector-specific peer-reviewed work cited

Coverage notes

Largest BizRadar opportunity by establishment count after retail. Includes: lawyers (5411*), accountants (541211, 541213), architects (541310), engineers (541330), IT consultants (541512), graphic designers (541430), photographers (541921), veterinarians (541940), management consultants (541611). All have strong review footprints. Sector-specific peer-reviewed review-economics work is a major research gap (flagged in Batch 9 follow-ups).

BIZRADAR PRIORITY

HIGH — top-2 BizRadar sector. Need ~15 sub-templates.

NAICS 55 — Management of Companies and Enterprises

Coverage class: THIN | 3 NAICS-6 codes | Priority: Defer permanently

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — covered
L3 — Batch 9 verified studies	Out of scope

Coverage notes

Holding companies, corporate offices, regional headquarters. Not consumer-facing. Out of scope.

BIZRADAR PRIORITY

Defer permanently — out-of-scope flag.

NAICS 56 — Administrative, Support & Waste Management

Coverage class: MODERATE | 44 NAICS-6 codes | Priority: HIGH

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 47% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

BizRadar-relevant: cleaning services (561720), pest control (561710), landscaping (561730), security services (561612), travel agencies (561510), tour operators (561520), event-planning (561920). Strong consumer-review footprint. ~15 of 44 codes in scope.

BIZRADAR PRIORITY

HIGH — local-services category, growing share of BizRadar customer requests.

NAICS 61 — Educational Services

Coverage class: MODERATE | 17 NAICS-6 codes | Priority: MEDIUM

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 56% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

BizRadar-relevant: tutoring services (611691), language schools (611630), driving schools (611692), fine-arts schools (611610), sports & recreation instruction (611620). ~7 of 17 in scope.

BIZRADAR PRIORITY

MEDIUM — moderate volume, predictable templates.

NAICS 62 — Health Care & Social Assistance

Coverage class: MODERATE | 39 NAICS-6 codes | Priority: HIGH

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — ~900K establishments
L2 — BLS BED 2013-cohort survival	BLS BED — 60% 5-yr survival (highest among major sectors — care continuity drives stability)
L3 — Batch 9 verified studies	Indirect via S015 (BrightLocal — review-reading applies); cross-industry CX (S017-S020); appointment SMS reminders (Batch 9 supplementary follow-up — Imperial College London 2008)

Coverage notes

Major BizRadar opportunity — high review presence on Google, Healthgrades, Vitals. In scope: dental offices (621210), chiropractors (621310), physical therapists (621340), home health (621610), child daycare (624410). Most other codes (hospitals, nursing facilities) out of scope. ~12 of 39 codes in scope.

BIZRADAR PRIORITY

HIGH — needs sector-specific verified-studies research in Batch 11.

NAICS 71 — Arts, Entertainment & Recreation

Coverage class: **STRONG** (gym sub-sector); **MODERATE** (rest) | 25 NAICS-6 codes | Priority: **HIGH** for non-gym recreation; **B**

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 47% 5-yr survival
L3 — Batch 9 verified studies	S008-S009 cover NAICS 713940 (gym) DIRECTLY; rest of sector cross-industry only

Coverage notes

Gym subsector (713940) is BATCH 9 strong-coverage. Remainder: golf courses (713910), bowling centers (713950), amusement parks (713110), arcades (713120), marinas (713930), independent artists (711510), performing-arts venues (711310). Roughly 15 of 25 in scope.

BIZRADAR PRIORITY

HIGH for non-gym recreation; Batch 9 already covers gym specifically.

NAICS 72 — Accommodation & Food Services

Coverage class: **STRONG** | 15 NAICS-6 codes | Priority: **DONE**

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — ~750K establishments
L2 — BLS BED 2013-cohort survival	BLS BED — 47% 5-yr (food); 51% (lodging)
L3 — Batch 9 verified studies	S001-S007 — DIRECT sector evidence (Cornell + Luca Harvard). 7 of 20 verified studies are NAICS-72 specific.

Coverage notes

BizRadar's flagship vertical. All 15 NAICS-6 codes in scope, all already covered through Batches 1-7. Batch 9 ground-truthed every customer-facing claim. This is the gold-standard reference for what 'STRONG coverage' looks like — the model the rest of the classifier expansion targets.

BIZRADAR PRIORITY

DONE — minor refresh for Batch 11; no new templates needed.

NAICS 81 — Other Services (except Public Admin)

Coverage class: MODERATE | 44 NAICS-6 codes | Priority: HIGH

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — 49% 5-yr survival
L3 — Batch 9 verified studies	Cross-industry only

Coverage notes

Personal-care services (812*) are critical: hair salons (812112), nail salons (812113), barber shops (812111), spas (812199). Auto repair (811111-811198) is also high-volume. Funeral services (812210), pet services (812910). ~25 of 44 codes in scope. Salon-specific Batch 9 follow-ups (Boulevard 2023, Etisia 2026 no-show research) supplement cross-industry studies.

BIZRADAR PRIORITY

HIGH — personal-care + auto-repair = top-5 BizRadar customer profile category.

NAICS 92 — Public Administration

Coverage class: GAP | 29 NAICS-6 codes | Priority: Defer permanently

Evidence layers

Layer	Status
L1 — Census CBP 2023 (TAM)	Census CBP 2023 — covered
L2 — BLS BED 2013-cohort survival	BLS BED — covered
L3 — Batch 9 verified studies	Out of scope

Coverage notes

Federal/state/local government. Out of scope by design — BizRadar doesn't serve government agencies as customers.

BIZRADAR PRIORITY

Defer permanently — out-of-scope flag.

Coverage Promotion Roadmap

What it would take to promote each MODERATE sector to STRONG (or each THIN sector to MODERATE). This is the work-list for Batch 11 sector polish and beyond.

MODERATE → STRONG promotions (Batch 11 priority)

Each row names a sector currently at MODERATE coverage and the kind of evidence that would promote it to STRONG.

Sector	Required evidence	Source path
NAICS 23 — Construction	Peer-reviewed work on residential-trade review-economics; or a CIRT/ENR benchmarking report with sector-specific review/customer-satisfaction figures.	Academic search (J. of Construction Engineering & Management; Construction Management & Economics); ENR Top 400 reports.
NAICS 44-45 — Retail Trade	Already STRONG via S014 (Womply Retail). Promotion to PEER-REVIEWED-STRONG would need an HBS/MIT-Sloan working paper on retail-review elasticity. Currently on Tier-3 vendor evidence.	Search for academic replication of Luca-Yelp findings in retail (Anderson, Magruder, Luca have related work).
NAICS 54 — Professional Services	Sector-specific studies on referrals economics for lawyers, accountants, IT consultants. The Schwab RIA 97% retention figure (Batch 9 follow-up) covers part of NAICS-52/523930 but not 54.	ABA Legal Marketing Reports; AICPA practice-management surveys; Vault/AdAge professional services research.
NAICS 56 — Administrative & Support	Sector-specific work on local-services review economics (cleaning, pest control, landscaping). Cross-industry vendor data exists (Angi, Thumbtack benchmarks) but is not peer-reviewed.	Angi annual reports; Thumbtack market data (vendor Tier-3 — already loosely cited).
NAICS 62 — Health Care	Studies on dental/specialty/urgent-care review impact. Healthgrades/Vitals/RateMDs publish aggregate data; academic literature exists (Yu et al on physician-rating websites) but uses outcomes other than revenue.	Health Affairs; J. of Medical Internet Research; Healthgrades benchmarking white papers.
NAICS 71 — Arts/Recreation (non-gym)	Sector-specific studies on golf/bowling/marina/arcade review economics. Largely a gap.	NGCOA/NGF for golf; BPAA for bowling; trade-association data.
NAICS 81 — Personal Services + Auto Repair	Sector-specific work on salon/spa/auto-repair review economics. Salon Boulevard data exists; auto-repair is an open gap. RepairPal aggregate data is starting point.	Boulevard 2023+ reports; RepairPal benchmarks; NADA/NAPA research.

THIN → MODERATE promotions (deferred)

Sectors currently THIN are mostly out of BizRadar's natural discovery surface. Promotion is possible but requires deciding to expand BizRadar's scope — not just doing more research.

- **NAICS 11 (Agriculture).** Direct-to-consumer farms (CSAs, U-pick orchards, farmstands) have local-discovery surface. A future scope expansion to 'agritourism' would promote roughly 8 of 64 NAICS-6 codes.
- **NAICS 31-33 (Manufacturing).** The consumer-facing edge — retail bakeries (311811), breweries (312120), wineries (312130), distilleries (312140), small-batch food (311941, 311991) — could be carved out as ~30 codes. Currently all go through their tasting-room/retail-front parent codes (312130 winery → 722410 drinking place when consumer-facing).
- **NAICS 48-49 (Transportation).** Local courier (492210), moving services (484210), and limo/taxi (485310, 485320, 485410) have review surface. A focused expansion ~10 codes.
- **NAICS 51 (Information).** Indie podcasters and content creators have brand-building footprints but not Google-Places-style ones. Out of natural BizRadar scope unless a separate creator-economy product is built.

WHAT 'DEFER PERMANENTLY' MEANS

Sectors marked Defer permanently in earlier chapters (Mining, Utilities, Management of Companies, Public Administration) should never have profile templates written. The classifier's out-of-scope filter should reject inputs from these sectors with a clear message: 'BizRadar is designed for consumer-facing local businesses. NAICS-21 / 22 / 55 / 92 inputs fall outside that scope.'

Layer 0 — Input Classifier Upgrade

The current production `server.js` classifies user input into one of 6 modes using brittle regex matching. Batch 10 specs an upgraded 10-mode classifier with explicit confidence scoring, ambiguity detection, and chain-vs-independent flagging.

Current state — `server.js` Layer 0 (v3)

Six modes, regex-only, with a generic location-fallback catching anything the classifier doesn't understand:

Mode	Detection rule
invalid	<code>len(input) < 3</code> — reject.
url	Contains <code>maps.google.com / yelp.com/biz / tripadvisor.com</code> — fetch the listing directly.
location	Has <code>' , '</code> pattern — geocode.
niche	Contains a known business-type keyword + comma — keyword + city search.
city	Has <code>' , '</code> pattern only — city-level search.
location_fallback	Anything else — try as a free-form geocode.

PROBLEMS WITH THE V3 CLASSIFIER

(1) The 18-keyword TYPES list misses ~95% of the Master Reference's actual business types — searches for 'lash bar', 'escape room', or 'food truck' fall through to `location_fallback`. (2) No chain detection — Marriott, Subway, Planet Fitness all get the same treatment as a neighborhood independent. (3) No ambiguity handling — 'bakery' could mean three different NAICS codes; the classifier just guesses. (4) No NAICS-direct input — power users can't pass a 6-digit code. (5) No `place_id` input — the Google Maps share-link case isn't handled.

Upgraded — Batch 10 spec, 10 modes

The new classifier expands to 10 modes. Key additions: place_id, brand_chain, naics_direct, and ambiguous (when a type-keyword matches multiple NAICS parents).

Mode	Detection rule	Change vs v3
invalid	len(input) < 3 OR matches blocklist (out-of-scope NAICS-21/22/55/92).	Same — reject early.
url	Contains a known directory hostname — fetch the listing directly.	Expand list: add g.co/kgs, share.google, yelp.to, fb.me, instagram.com/, tiktok.com/@, doordash.com/store, ubereats.com/store.
place_id	NEW — explicit Google place_id input (ChIJ...)	Direct dispatch to Google Places Details API.
brand_chain	NEW — name matches a top-500 chain (Marriott, Subway, Planet Fitness, etc.)	Special handling: chain rules apply (e.g. Luca finding does NOT apply to chains; chain templates differ from independent templates).
naics_direct	NEW — input is a 6-digit NAICS code itself.	Direct lookup; useful for power users and the SBA-import workflow.
location	Has ' ' AND city/state.	Same; tightened regex; requires a comma OR clearly-formatted address.
niche_typed	Contains a known type keyword (expanded from 18 → ~250 keywords from Part-2 catalog).	Same shape; much larger keyword table. Add per-keyword confidence (e.g. 'sushi' = 0.95, 'shop' = 0.30). Tag with NAICS-6 candidate set.
ambiguous	NEW — input matches >1 type keyword and the candidates have different NAICS parents.	Trigger disambiguation prompt: ask user 'Did you mean (a) bakery — retail; (b) bakery — coffee shop; (c) bakery — wholesale?'
city	Has ', ' pattern only.	Same — city-level search.
free_form_geocode	Anything else — last resort.	Same; mark response confidence 'LOW'.

Confidence scoring

Each detected mode produces a confidence score, returned to downstream layers as part of the classifier output. Low-confidence outputs trigger explicit user clarification rather than silently producing a low-quality report.

Confidence	Meaning	Downstream behavior
HIGH	Pattern match unambiguous (URL, place_id, NAICS code, exact-match keyword)	Proceed to Layer 1 directly.
MEDIUM	Type-keyword match with one strong NAICS candidate	Proceed; tag report 'auto-detected: '.
LOW	Free-form geocode or ambiguous input	Surface to user: 'I'm interpreting this as ; is that right?'

Disambiguation Rules

Six common ambiguous queries that the v3 classifier silently guessed on. Each gets a deterministic tiebreak rule.

The Layer 0 classifier should NEVER guess between equally-valid NAICS candidates. It should either apply a deterministic rule (when one exists) or ask the user. The six examples below are the most common ambiguous queries — these rules are pulled into the registry for Batch 12 implementation.

Query: 'bakery'

Candidate NAICS codes

NAICS	Label	When this fits
311811	Retail Bakeries (manufacturing-attached)	Has on-site oven/kitchen; sells direct from the manufacturing space. Donut shop, cupcake shop, bagel shop.
722515	Snack and Nonalcoholic Beverage Bars	Buys baked goods or assembles from prepared components. Coffee-shop-with-pastries pattern.
445291	Baked Goods Retailers	Pure retail — no on-site production. Distinct from 311811.

TIEBREAK RULE

Default to 311811 unless query explicitly contains 'cafe', 'coffee', 'shop' (without 'bake-' prefix). Sub-cluster scoring applies: presence of espresso machine in Google Places photos = 722515; presence of large oven/baking equipment = 311811.

Query: 'spa'

Candidate NAICS codes

NAICS	Label	When this fits
812199	Other Personal Care Services	Day spas, med spas, traditional massage spas. The default.
721110	Hotels (except Casino Hotels) and Motels	Wellness/destination retreats with overnight stays — 'wellness retreat' or 'destination spa'.
621399	Offices of All Other Misc Health Practitioners	Med spas with licensed-medical-staff procedures (Botox, fillers, laser). Ambiguous — many med spas operate under 812199 even with medical staff.

TIEBREAK RULE

Default to 812199. Shift to 721110 if query contains 'retreat', 'lodge', 'overnight', or any night-stay signal. Shift to 621399 if query contains 'medical', 'aesthetic medicine', 'plastic surgery'.

Query: 'studio'

Candidate NAICS codes

NAICS	Label	When this fits
611620	Sports & Recreation Instruction (yoga, pilates studios)	Group fitness/movement instruction.
541921	Photography Studios, Portrait	Portrait photography studios.
541490	Other Specialized Design Services (design studios)	Architecture/design studios.
711510	Independent Artists, Writers, and Performers (artist studios)	Recording studios, art studios.
711410	Agents and Managers for Artists	Talent agency studio — rare.

TIEBREAK RULE

Use modifier matching. 'yoga studio' → 611620; 'photo studio' → 541921; 'recording studio' → 711510; 'design studio' → 541490. If no modifier, ask the user — 'studio' alone is too ambiguous for a deterministic mapping.

Query: 'salon'

Candidate NAICS codes

NAICS	Label	When this fits
812112	Beauty Salons (full-service hair + skin + nail)	The default for any 'salon' input.
812113	Nail Salons	If 'nail' explicitly in query.
812111	Barber Shops	If 'barber' or 'men' in query — note BizRadar already has this template.

TIEBREAK RULE

Default to 812112. Shift to 812113 only if 'nail' present. Shift to 812111 only if 'barber' present. The salon→barber split is one of the most common mis-classifications in the v3 server.js logic; Batch 10 spec fixes it.

Query: 'shop'

Candidate NAICS codes

NAICS	Label	When this fits
Various 44-45	Retail Trade — depends on goods sold	Bookshop → 459210; clothing shop → 458110; coffee shop → 722515; barbershop → 812111. 'Shop' alone is too generic.

TIEBREAK RULE

REJECT 'shop' as a single keyword. Always require a modifier. The classifier should respond with 'Could you tell me what kind of shop?' if 'shop' is the only type-keyword detected.

Query: 'agency'

Candidate NAICS codes

NAICS	Label	When this fits
541810	Advertising Agencies	If 'ad' or 'marketing' present.
541870	Advertising Material Distribution Services	Niche.
561510	Travel Agencies	If 'travel' present.
561311	Employment Placement Agencies	If 'staffing', 'recruiting' present.
541612	Human Resources Consulting Services	If 'HR' present.

TIEBREAK RULE

Always require a modifier. 'agency' alone triggers a clarification prompt.

WHAT LIVES IN THE REGISTRY (BATCH 12)

These six are the worked examples. The full disambiguation ruleset — every business-type keyword that maps to multiple NAICS parents — ships as a JSON registry in Batch 12. Estimated rule count: ~60-80 disambiguation entries.

Part 2 — Modern Formats Catalog

The Master Reference's Part 2 lists 452 common business types that don't have their own NAICS-6 code but show up frequently in BizRadar customer queries (food trucks, ghost kitchens, lash bars, escape rooms, etc.). This section maps them to BizRadar's coverage classes and to the Layer 0 keyword-recognition table.

Category	# types	Primary NAICS parents	Scope
Food Service (modern formats)	42	722330, 722513, 722515, 722511, 722410, 311811	IN SCOPE
Lodging & Hospitality (modern formats)	14	721110, 721199, 721211, 721191, 721310	IN SCOPE
Retail & E-commerce (modern formats)	42	459999, 455110, 454110, 454390, 459110-459930	IN SCOPE
Personal Services (modern formats)	43	812199, 812113, 812112, 812191	IN SCOPE
Fitness, Wellness & Recreation	53	713940, 812199, 711211, 713990, 611620	IN SCOPE
Tech, Digital & Creative	50	541512, 541810, 541921, 711510, 519290	PARTIAL
Trades & Home Services	59	23-* (Construction sub-codes)	IN SCOPE
Education & Tutoring (modern formats)	33	611691, 611620, 611610, 611630, 611699	IN SCOPE
Healthcare & Medical (modern formats)	42	621* (most), 446191, 812199	IN SCOPE
Professional Services (modern formats)	36	5411* (legal), 5412* (accounting), 5414* (design)	IN SCOPE
Niche & Emerging	38	Various	PARTIAL

Food Service (modern formats) — 42 types

Scope: IN SCOPE | Maps to NAICS: 722330, 722513, 722515, 722511, 722410, 311811

Food trucks, ghost kitchens, pop-ups, boba shops, juice bars, ice cream shops, donut shops, taco trucks, ramen shops, hot pot, sushi bars, wine bars, sports bars. All roll up to NAICS-72 parents covered by Batches 5-9 (strong coverage). Layer 0 classifier needs explicit rules for 'food truck', 'ghost kitchen', 'pop-up' as they share parent codes with permanent restaurants but have different unit economics.

Lodging & Hospitality (modern formats) — 14 types

Scope: IN SCOPE | Maps to NAICS: 721110, 721199, 721211, 721191, 721310

Boutique hotels, glamping, Airbnb hosts, vacation rental management, hostels, B&Bs, eco-lodges, wellness retreats. STR (short-term rental) hosts are a fast-growing customer segment. Roll up to NAICS-72 parents — strong evidence layer. Classifier needs 'Airbnb', 'STR', 'vacation rental', 'glamping' keyword-recognition and a 721199 → 'all other traveler accommodation' fallback chain.

Retail & E-commerce (modern formats) — 42 types

Scope: IN SCOPE | Maps to NAICS: 459999, 455110, 454110, 454390, 459110-459930

Pop-up shops, online-only DTC brands, Etsy/handmade sellers, antique malls, pet boutiques, plant shops, vintage clothing, sneaker resale, vinyl shops. Roll up to NAICS-44/45 parents — Batch 9 STRONG via S014 (Womply Retail).

Personal Services (modern formats) — 43 types

Scope: IN SCOPE | Maps to NAICS: 812199, 812113, 812112, 812191

Lash extensions, microblading, eyebrow threading, med spas, IV therapy, cryotherapy, lash bars, brow bars, Brazilian wax, men's grooming lounges. NAICS-81 moderate coverage; Batch 11 should add specialized templates for the highest-volume modern formats (med spa, lash bar).

Fitness, Wellness & Recreation — 53 types

Scope: IN SCOPE | Maps to NAICS: 713940, 812199, 711211, 713990, 611620

Boutique studios (Pilates, barre, spin, yoga), CrossFit boxes, climbing gyms, F45/Orangetheory-style HIIT, recovery studios (cryo, sauna, contrast), youth sports academies, escape rooms, axe-throwing, trampoline parks. 713940 (gym) STRONG via Batch 9; rest moderate.

Tech, Digital & Creative — 50 types

Scope: PARTIAL | Maps to NAICS: 541512, 541810, 541921, 711510, 519290

Web designers, app developers, podcast studios, video production, social-media-management agencies, content creators, photographers, voice-over artists. Many are sole-proprietors without local-discovery footprint. ~25 of 50 in scope.

Trades & Home Services — 59 types

Scope: IN SCOPE | Maps to NAICS: 23-* (Construction sub-codes)

Plumbers, electricians, HVAC, roofers, painters, landscapers, junk removal, gutter cleaning, pressure washing, holiday-light installers, mobile auto-detailing, locksmiths. NAICS-23 (or 81 for auto). Strong consumer-review presence on Google/Yelp/Angi. Top-3 BizRadar opportunity.

Education & Tutoring (modern formats) — 33 types

Scope: IN SCOPE | Maps to NAICS: 611691, 611620, 611610, 611630, 611699

Test-prep tutors, music schools, language schools, coding bootcamps, art schools, dance schools, academic-coaching services, special-needs tutoring. Moderate coverage.

Healthcare & Medical (modern formats) — 42 types

Scope: IN SCOPE | Maps to NAICS: 621* (most), 446191, 812199

Concierge medicine, telehealth-first practices, med spas, IV therapy clinics, weight-loss clinics, ketamine clinics, hormone clinics, urgent-care centers, dental specialty (orthodontics, endodontics). Major BizRadar opportunity. Need sector-specific verified studies in Batch 11 — gap flagged.

Professional Services (modern formats) — 36 types

Scope: IN SCOPE | Maps to NAICS: 5411* (legal), 5412* (accounting), 5414* (design)

Estate planners, immigration lawyers, fractional CFO services, bookkeeping (online-first), virtual paralegals, brand strategy consultants, executive coaches, fractional CMO services. NAICS-54 moderate coverage; sector-specific studies a gap.

Niche & Emerging — 38 types

Scope: PARTIAL | Maps to NAICS: Various

Subscription newsletters, Patreon creators, NFT artists, day traders, vending machine routes, ATM operators, self-storage, laundromats, drone-light-show operators, mobile billboard. Mostly out of Layer 0 scope unless they have a physical-discovery footprint (laundromats, self-storage YES; Patreon creators NO).

TRANSLATION TO LAYER 0

Every IN SCOPE Part-2 entry contributes to the Layer 0 keyword table. The v3 classifier had 18 keywords; Batch 10 specs ~250 (every Part-2 entry that's IN SCOPE, plus the v3 18). Each keyword carries: the source phrase, the primary NAICS-6 candidate, and a confidence score. Multi-candidate keywords route through the disambiguation rules from the previous section.

Out-of-Scope Filter

BizRadar refuses inputs from these sectors with a clear message — silently producing a low-quality report is worse than refusing.

NAICS-2 sectors out of scope

NAICS-2	Sector	Reason out of scope
21	Mining, Quarrying, Oil & Gas	B2B/heavy-industry; no local-discovery surface.
22	Utilities (regulated)	Regulated utilities; not BizRadar's customer base.
55	Management of Companies & Enterprises	Holding companies, regional HQs; not consumer-facing.
92	Public Administration	Government agencies; not BizRadar customers by design.

NAICS-6 patterns out of scope (within in-scope NAICS-2)

Even within in-scope NAICS-2 sectors, certain code ranges are out of scope. The most important: NAICS 31-33 manufacturing (346 codes — only ~30 are consumer-facing).

Pattern	Description	Notes
311111-311999	Most manufacturing	B2B production; exception list applies (311811 retail bakery, 312120 brewery, 312130 winery, 312140 distillery).
481*-486*	Air, rail, water, pipeline transport	B2B / regulated.
511*-518*	Most information sub-sectors	Software publishers, broadcasting, ISPs not local-discovery.
521*-525*	Banks, securities, funds	Regulated finance; not BizRadar fit.
531110, 531120, 531190	Lessors of real estate (large)	Commercial leasing; B2B.
813*	Membership organizations, religious orgs	Not commercial businesses in BizRadar's sense.

USER-FACING MESSAGE — EXAMPLE

'Your input maps to NAICS 211130 (Crude Petroleum Extraction). BizRadar is designed for consumer-facing local businesses — industries with reviews on Google Maps, Yelp, or similar local-discovery surfaces. NAICS-21 (Mining/Extraction) doesn't fit that pattern. Did you mean to enter a different business?'

Chain vs Independent — Handling

Batch 9's most-cited finding (S006 — Luca/Harvard 2016) is that Yelp star ratings move revenue 5-9% per star — but ONLY for INDEPENDENT restaurants. Chain restaurants show no significant effect. This finding has implications across BizRadar's whole review-management narrative — and requires the classifier to detect chains explicitly.

Why this matters

- **Wrong magnitude estimates.** Applying the 5-9% Yelp elasticity to a Subway franchisee massively overstates the expected revenue lift from review improvement. The customer report would be inaccurate.
- **Different recommended actions.** A chain franchisee's review-management options are constrained by corporate brand guidelines. The recommended actions differ from an independent's.
- **Different competitive set.** A McDonald's competes with other QSR chains (Burger King, Wendy's). Comparing it to the local independent burger joint produces misleading competitive analysis.

Top-100 chains — handling table (sample)

The full top-100 list ships in the registry (Batch 12). Sample entries:

Brand	NAICS	Label	Treatment
McDonald's	722513	Limited-Service Restaurants	Use chain-specific template; do NOT apply Luca finding (chain Yelp ratings do not move revenue per S006).
Subway	722513	Limited-Service Restaurants	Same as above.
Starbucks	722515	Snack and Nonalcoholic Beverage Bars	Same — chain template, not independent.
Marriott / Hilton / Hyatt branded	721110	Hotels	Apply Cornell hotel templates (S001-S004) but flag as chain — scale the effect estimates down by ~30% per chain-vs-independent implication of S006 logic.
Planet Fitness / LA Fitness / Anytime Fitness	713940	Gym	Standard gym template (Batch 9 S008-S009) applies; chain-specific retention may differ from industry 66.4% — flag for verification.
Great Clips / SuperCuts	812112	Beauty Salons	Chain template; salon-walk-in model. Boulevard 78%/39% online-vs-walk-in retention finding does NOT apply (chain operates walk-in by design).

DETECTION MECHANIC

The classifier matches the input business name (or Google Places result name) against the top-100 chain list. Match → chain template; no match → assume independent. Top-100 covers ~80% of chain-establishment volume in the U.S.; the long tail of regional chains is acceptable false-negative (report defaults to independent template, slight overstatement of review-management impact).

Appendix A — Master Coverage Matrix

Single-page reference: every NAICS-2 sector with its establishment count, survival rate, Batch 9 study coverage, and Batch 11 priority. This is the working document for Batch 11 sector polish.

NAICS	Sector	# codes	Surv.	Class	Priority
11	Agriculture, Forestry, Fishing & Hu...	64	BLS BED — 51%	MODERATE	Low
21	Mining, Quarrying, Oil & Gas Extrac...	21	BLS BED — 56%	THIN	Defer
22	Utilities	14	BLS BED — 65%	THIN	Defer
23	Construction	31	BLS BED — 36%	MODERATE	HIGH
31-33	Manufacturing	346	BLS BED — 49%	THIN	MEDIUM
42	Wholesale Trade	69	BLS BED — 53%	THIN	Defer
44-45	Retail Trade	57	BLS BED — 47%	STRONG	HIGH
48-49	Transportation & Warehousing	57	BLS BED — 47%	MODERATE	MEDIUM
51	Information	29	BLS BED — 49%	THIN	Defer
52	Finance & Insurance	35	BLS BED — 51%	MODERATE	MEDIUM
53	Real Estate, Rental & Leasing	24	BLS BED — 50%	MODERATE	MEDIUM
54	Professional, Scientific & Technica...	49	BLS BED — 51%	MODERATE	HIGH
55	Management of Companies and Enterpr...	3	BLS BED — covered	THIN	Defer permanen
56	Administrative, Support & Waste Man...	44	BLS BED — 47%	MODERATE	HIGH
61	Educational Services	17	BLS BED — 56%	MODERATE	MEDIUM
62	Health Care & Social Assistance	39	BLS BED — 60%	MODERATE	HIGH
71	Arts, Entertainment & Recreation	25	BLS BED — 47%	STRONG	HIGH for non-g
72	Accommodation & Food Services	15	BLS BED — 47%	STRONG	DONE
81	Other Services (except Public Admin...	44	BLS BED — 49%	MODERATE	HIGH
92	Public Administration	29	BLS BED — covered	GAP	Defer permanen

Aggregate coverage statistics

Metric	Value
NAICS-2 sectors total	20
NAICS-6 codes total (Part 1)	1,012
Part-2 modern format entries	452
Master Reference total	1,464
NAICS-6 codes in STRONG sectors	97
NAICS-6 codes in MODERATE sectors	404
NAICS-6 codes in THIN sectors	482
NAICS-6 codes in GAP sectors	29
NAICS-2 marked HIGH priority for Batch 11	7
NAICS-2 marked Defer permanently	6

Appendix B — classifierRegistry.json Spec

Batch 12 ships the production classifier as a JSON registry. This appendix specifies the schema.

Schema overview

```
classifierRegistry.json +- version: "10.0" +- generated: "2026-05-05" +- outOfScope: | +- naics2:
["21","22","55","92"] | +- naics6Patterns: ["481*", "482*", ...] +- chains: | +- [{ name, naics6, label,
treatment }, ...] # top-100 +- keywords: | +- [{ phrase, naics6, confidence }, ...] # ~250 +- disambiguation:
| +- { "": { | candidates: [{ naics6, label, when }], | tiebreakRule: "", | askUser: true|false | }, ... } #
~60-80 entries +- modernFormats: | +- [{ format, parentNaics6, scope, ... }, ...] # 452 entries +- sectors: +-
{ "11": { label, l1, l2, l3Studies, class, priority }, ... }
```

Lookup flow

Layer 0 receives the raw user input. The classifier walks the registry in this order:

1. URL match — directory hostname → fetch listing directly.
2. place_id detection — ChIJ-prefix → Google Places Details API.
3. NAICS-direct — 6-digit numeric input → registry.sectors lookup.
4. Chain match — fuzzy match against registry.chains → chain template flag set.
5. Keyword match — every Part-2 modern format and v3 keyword checked against registry.keywords. If multi-match → step 6; else proceed.
6. Disambiguation — if multiple NAICS candidates, apply registry.disambiguation tiebreak rule. If askUser=true, prompt the user.
7. Out-of-scope check — if resolved NAICS is in registry.outOfScope, return 'out of scope' message.
8. Geocode fallback — if no keyword/chain/place_id match but input has address-like structure, free-form geocode with LOW confidence flag.

Versioning

The registry is versioned. Major version bumps when the schema changes; minor version bumps when content changes (new keywords, new chains, etc.). Production server.js reads the registry version at startup and refuses to run if the version is older than its expected minimum.

WHY JSON, NOT HARD-CODED CONSTANTS

The classifier registry is the most-frequently-updated piece of BizRadar data. Chains rebrand (Dunkin' Donuts → Dunkin'), new modern formats emerge (smashburger pop-ups, NA-cocktail bars), keyword false-positives surface in production logs. JSON makes these updates a content change, not a code change.

Appendix C — Follow-ups for Batches 11 & 12

What Batch 10 explicitly defers to later batches.

Batch 11 — Sector Polish (~70 pages)

1. Profile templates for HIGH-priority MODERATE sectors. Construction (NAICS-23), Health Care (NAICS-62), Professional Services (NAICS-54), Personal Services + Auto Repair (NAICS-81), Administrative Services (NAICS-56), Arts/Recreation non-gym (NAICS-71). Estimated 30-40 templates total.
2. Sector-specific verified-studies expansion. The MODERATE→STRONG promotion roadmap from Part 2 names the search paths: construction-management literature, health-affairs journals, ABA legal-marketing reports, Boulevard salon data, etc. Successful research adds new study IDs (S021+) to the Batch 9 registry.
3. Top-100 chain list completion. Batch 10 ships ~6 sample entries; Batch 11 fills out the full 100.
4. Disambiguation ruleset completion. Batch 10 ships 6 worked examples; Batch 11 fills out the estimated 60-80 entries by auditing production query logs from server.js.

Batch 12 — Final Integration (~30 pages)

1. Ship classifierRegistry.json. Schema specced in Appendix B.
2. Ship verifiedStudies.json. Schema specced in Batch 9 registry-spec section.
3. Update server.js Layer 0 to consume both registries. Replace the 6-mode regex chain with the 10-mode pipeline.
4. Citation linter. Customer-facing report renderer fails any claim that doesn't reference a study ID from verifiedStudies.json. This is the production enforcement of Batch 9's citation policy.
5. Out-of-scope flag in user-facing UI. Refuses NAICS-21/22/55/92 inputs with the specced message.
6. End-to-end integration test plan. Ten representative queries (URL, place_id, NAICS direct, chain, ambiguous, out-of-scope, etc.) with expected classifier output.

Deferred indefinitely (out of v4 scope)

- Multi-language input handling (Spanish business-type keywords, e.g. 'panadería'). Batch 10's keyword table is English-only.
- International NAICS-equivalent classifiers (Canadian NAICS, Mexican SCIAN, EU NACE). v4 is U.S.-only.
- B2B-specific BizRadar product expansion (would unlock NAICS-21, 22, 42, 51, most of 31-33). Separate product decision, not a spec gap.
- Real-time chain updates (mergers, rebrands). Currently depends on manual registry updates. Could be automated via OpenCorporates API in v5.

Closing — Batch 10 Summary

Batch 10 delivered: 100% NAICS-2 coverage taxonomy, 20 sector deep dives, Layer 0 classifier upgrade spec, 6 disambiguation rules, Part-2 modern formats catalog, out-of-scope filter, chain handling, registry spec, follow-up roadmap.

What this batch ships into the spec

- **Coverage taxonomy.** Every NAICS-2 sector classified STRONG / MODERATE / THIN / GAP. Every Part-2 modern format scoped IN / PARTIAL / OUT.
- **Layer 0 upgrade.** 6 → 10 modes. Adds place_id, brand_chain, naics_direct, ambiguous. Confidence scoring explicit.
- **Disambiguation.** 6 worked examples; full ruleset deferred to Batch 11.
- **Out-of-scope filter.** 4 NAICS-2 sectors and 6 NAICS-6 patterns named.
- **Chain handling.** Top-100 sample entries; full list deferred to Batch 11.
- **Registry spec.** classifierRegistry.json schema for Batch 12 implementation.

Quality scorecard — after Batch 10

Dimension	Pre-B10	Post-B10	B11 target	B12 target
TAM accuracy (Census CBP)	90	90	90	92
Failure-rate accuracy (BLS)	92-95	92-95	94	95
Source citability	90+	90+	92	95
Narrative grounding	92	92	94	95
Classifier coverage	~40%	100%	100%	100%
Disambiguation explicit	0	Spec	Full ruleset	Production
Chain detection	None	Spec	Full top-100	Production
Out-of-scope handling	None	Spec	Spec	Production

Two batches remaining

Batch 11 (sector polish, ~70 pages) takes the HIGH-priority MODERATE sectors and writes the profile templates plus sector-specific verified-studies research. Batch 12 (final integration, ~30 pages) ships the production registries and wires the Layer 0 upgrade into server.js.

After Batch 12, BizRadar v4 is feature-complete at the spec level. The remaining 5 points to 100/100 require user-driven work outside the spec layer: live A/B testing of recommendations, commissioned research for the deepest gap sectors (Construction, Health Care, Professional Services), legal review for compliance-sensitive sectors, and customer-interview validation of the report-template language.

SPEC FOOTPRINT CHECK

The Batch 8 + 9 + 10 trilogy establishes BizRadar's spec footprint. Batch 8 = data layer (TAM, failure rates). Batch 9 = evidence layer (verified studies). Batch 10 = classifier layer (which businesses get which evidence). Together they cover the three structural questions a customer-facing report answers: how big is this market, how risky is it, and what specifically should the operator do?